

MASTER PROMPT — Build "Nastor AI Studio"

Overview

You are an expert software architect, senior full-stack engineer, AI engineer, music producer, audio DSP engineer, UX/UI designer, and DevOps engineer.

Your task is to build a production-ready desktop and web application called **Nastor AI Studio**.

This is not just another AI chatbot.

It is an **AI-powered Music Production Operating System** designed specifically for producers, composers, DJs, songwriters, recording engineers, and record labels.

The application should function as an intelligent music production assistant that helps users from the first idea to the final mastered release.

The project must use modern architecture, clean code, scalability, security, and modular development.

Tech Stack

Frontend

- Next.js
- React
- TypeScript
- Tailwind CSS
- Framer Motion
- Electron (Desktop)
- Zustand
- React Query
- ShadCN UI

Backend

- Node.js
- NestJS
- PostgreSQL
- Prisma ORM
- Redis
- WebSockets
- Docker
- REST + GraphQL APIs

Authentication

- Google Login
- GitHub Login
- Email Authentication
- OAuth
- JWT

Cloud

- AWS / Vercel
- Supabase Storage
- Cloudflare

Design Style

The interface should feel like a premium creative application inspired by:

- Linear
- Notion
- Arc Browser
- Ableton Live
- FL Studio
- Spotify
- Adobe Creative Cloud

Requirements:

- Beautiful dark mode
- Glassmorphism
- Neon accents
- Smooth animations
- Dockable panels
- Customizable workspace
- Keyboard shortcuts
- Multi-monitor support
- Responsive design

Dashboard

The dashboard should include:

- AI Assistant

- Recent Projects
 - Music Library
 - Templates
 - Samples
 - Reference Tracks
 - AI Activity Feed
 - Recent Exports
 - Plugin Manager
 - Collaboration Panel
 - Mastering Queue
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AI Music Assistant

The AI assistant should:

- Chat naturally
 - Understand music production terminology
 - Understand Afro House
 - Understand Amapiano
 - Understand Deep House
 - Understand Tech House
 - Understand Pop
 - Understand Hip Hop
 - Understand EDM
 - Remember project context
 - Suggest production improvements
 - Explain mixing decisions
 - Explain mastering decisions
 - Provide songwriting assistance
 - Generate production ideas
 - Suggest arrangements
 - Act like a professional producer
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Lyrics Studio

Allow users to:

- Generate lyrics
- Rewrite lyrics
- Improve lyrics
- Translate lyrics
- Generate hooks
- Generate choruses

- Generate verses
 - Generate bridges
 - Create harmonies
 - Detect rhyme schemes
 - Detect syllables
 - Match lyrics to BPM
 - Match lyrics to emotion
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Prompt Studio

Allow users to:

- Save prompts
 - Organize prompts
 - Share prompts
 - Generate prompts
 - Improve prompts
 - Build prompt templates
 - Prompt history
 - Prompt versioning
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AI Music Generation

Integrate providers using a modular architecture:

- Suno API
- Future OpenAI audio models
- Google AI
- Anthropic
- Additional music APIs

Users should be able to generate:

- Songs
 - Instrumentals
 - Loops
 - Chord progressions
 - Basslines
 - Drum grooves
 - Melodies
 - Vocal ideas
 - Harmonies
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Arrangement Engine

Create an AI arrangement assistant capable of:

- Building complete song structures
 - Intro generation
 - Verse generation
 - Chorus generation
 - Drop generation
 - Build-up generation
 - Bridge generation
 - Outro generation
 - Automatic song extension
 - Loop expansion into full arrangements
 - Intelligent transitions
 - Energy curve analysis
 - Arrangement suggestions from uploaded references (without copying copyrighted material)
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Reference Track Intelligence

Users should be able to upload:

- WAV
- MP3
- AIFF
- FLAC
- MIDI

The AI should analyze:

- BPM
- Musical key
- Chord progression
- Mood
- Energy
- Groove
- Instrumentation
- Song structure
- Frequency balance
- Loudness
- Stereo image
- Dynamic range
- Vocal style
- Drum characteristics

The AI should then suggest:

- Better arrangement ideas
- Mix improvements
- Sound design inspiration
- Similar production techniques
- Chord suggestions
- Tempo recommendations
- Transition ideas

The AI must **never clone or reproduce copyrighted material**. It should only extract musical characteristics and provide original suggestions.

Stem Separation

Include AI stem separation.

Separate:

- Vocals
- Drums
- Bass
- Melody
- Pads
- FX
- Piano
- Guitar
- Other instruments

Allow exporting each stem individually.

MIDI Generator

Generate MIDI for:

- Piano
- Bass
- Drums
- Chords
- Strings
- Plucks
- Leads
- Arpeggios

Allow drag-and-drop export into any DAW.

DAW Integration Layer

Create a modular integration system for:

- FL Studio
- Ableton Live
- Cubase
- Logic Pro
- Reaper
- Studio One
- Bitwig

Features:

- MIDI export
- Audio export
- Project synchronization (where supported)
- Plugin communication (future)
- Python scripting integration where available
- Session analysis
- Smart production recommendations

The architecture should make it easy to add future DAW integrations.

Sample & Loop Manager

Users can:

- Upload samples
 - Organize libraries
 - Tag samples
 - Search with AI
 - Detect key
 - Detect BPM
 - Auto-categorize
 - Recommend compatible samples
 - Find similar sounds
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AI Mixing Assistant

Features:

- Frequency analysis
 - Stereo analysis
 - Dynamic analysis
 - EQ suggestions
 - Compression suggestions
 - Reverb suggestions
 - Delay suggestions
 - Gain staging assistant
 - Loudness assistant
 - Vocal chain suggestions
 - Drum bus suggestions
 - Master bus suggestions
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AI Mastering Hub

Integrate a mastering workflow inspired by services such as LANDR and similar platforms where APIs or export workflows are available.

Users should be able to:

- Upload a reference master
- Compare A/B masters
- Choose mastering styles
- Club Master
- Streaming Master
- Afro House Master
- Amapiano Master
- Radio Master
- Vinyl Master
- Loud Master
- Transparent Master

Provide:

- LUFS analysis
- True Peak analysis
- Dynamic range analysis
- Stereo width analysis
- EQ curve comparison
- Match tonal balance to uploaded references without copying copyrighted audio

Export Center

Generate:

- Radio Edit
- Extended Mix
- Club Mix
- DJ Intro
- DJ Outro
- Instrumental
- Acapella
- Performance Mix
- Streaming Version
- Demo Version
- Stems
- MIDI
- WAV
- MP3
- FLAC

Support export presets.

Project Manager

Include:

- Auto Save
 - Version History
 - Project Snapshots
 - Undo History
 - Cloud Sync
 - Team Collaboration
 - Comments
 - Project Templates
 - Smart Search
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AI Memory

Remember:

- Favorite genres

- Favorite plugins
- Preferred BPM
- Favorite scales
- Mixing style
- Arrangement style
- Vocal style
- Preferred mastering settings
- Frequently used prompts

Allow users to import their identity and preferences using:

- Google Account
- GitHub Account

Build a secure profile system that learns user preferences over time while giving users full control over their data.

Plugin Marketplace

Future marketplace for:

- AI plugins
 - Prompt packs
 - Sample packs
 - Presets
 - Templates
 - MIDI packs
 - Effect chains
 - Community tools
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Performance

Target:

- Fast startup
- Lazy loading
- Background processing
- GPU acceleration where available
- Offline support for compatible features
- Secure API key management
- Modular plugin architecture
- Automated testing
- CI/CD pipeline

- Comprehensive documentation
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Folder Structure

Use enterprise-level architecture with clear separation of concerns:

- /apps
 - /packages
 - /backend
 - /frontend
 - /services
 - /plugins
 - /ai
 - /audio
 - /integrations
 - /shared
 - /database
 - /docs
 - /tests
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Development Roadmap

Phase 1 (MVP)

- Authentication
- Dashboard
- AI Chat
- Lyrics Studio
- Prompt Studio
- Suno integration
- Project Manager
- Reference Track Upload
- Basic Analysis

Phase 2

- Arrangement Engine
- Stem Separation
- MIDI Generation
- Sample Manager
- AI Mixing Assistant

Phase 3

- AI Mastering Hub
- DAW Integrations
- Collaboration
- Marketplace
- Cloud Sync

Phase 4

- Mobile Companion App
 - VST/AU Plugin
 - Real-time DAW Assistant
 - Voice Commands
 - AI Automation Workflows
 - Label Management Tools
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Final Goal

Create the world's most advanced AI-powered music production platform—an intelligent creative workspace that assists producers from the initial idea through songwriting, composition, arrangement, sound selection, mixing, mastering, version creation, and release preparation.

The application must prioritize originality, user creativity, and copyright compliance while delivering a premium, extensible, enterprise-grade architecture.